

JEE MAIN REVISION MASTER: MOTION IN A PLANE 2026 1 QUESTIONS 1.67% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Motion In A Plane - Core Application	9 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Motion In A Plane - Core Application (High Yield — 9 PYQs)

Core Principle: Key problem pattern in Kcet Motion In A Plane.

■■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Two bodies are projected with the same velocity. If one is projected at an angle of 30° and the other at 45° to the horizo..." [Ans: B]
- [2025 Shift] Q#2 (MCQ): "In the projectile motion of a particle on a level ground, which of the following remains constant with reference to time and position?..." [Ans: N/A]
- [2022 Shift] Q#3 (MCQ): "Two objects are projected at an angle θ° and $(90-\theta^\circ)$, to the horizontal with the same speed. The ratio of their..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Two bodies are projected with the same velocity. If one is projected at an angle of 30° and the other at 45° to the horizontal, then the ratio of maximum heights attained is

- (A) 3:1
- (B) 1:2 **[Correct]**
- (C) 4:1
- (D) 1:3

Correct Answer: B

Solution Analysis: For a projectile, the maximum height is given by $H = \frac{u^2 \sin^2 \theta}{2g}$. Here both bodies are projected with the same velocity, so u is the same for both. Also, g is the same. So maximum height depends only on $\sin^2 \theta$. For the first body, $\theta = 30^\circ$, $H_1 = \frac{u^2 \sin^2 30^\circ}{2g}$. For the second body, $\theta = 45^\circ$, $H_2 = \frac{u^2 \sin^2 45^\circ}{2g}$. So, $\frac{H_1}{H_2} = \frac{\sin^2 30^\circ}{\sin^2 45^\circ}$. We know, $\sin 30^\circ = \frac{1}{2}$ and $\sin 45^\circ = \frac{1}{\sqrt{2}}$. So, $\frac{H_1}{H_2} = \frac{(\frac{1}{2})^2}{(\frac{1}{\sqrt{2}})^2} = \frac{1/4}{1/2} = \frac{1}{2}$.